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[Rubric]

Element E

Scientific principles and considerations

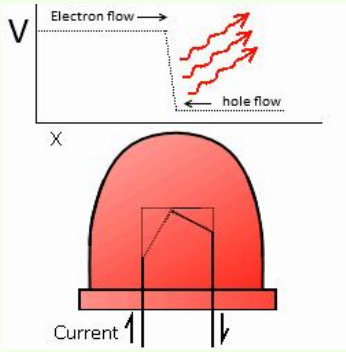
Some important technical considerations to make while creating our design and adjusting it to fit the project's needs include aspects of engineering, science and technology. In other words, there are many things to focus on in terms of how According to the NIH, different colored lights can have varying effects on the mood and productivity of an individual. Generally, it was found through many studies that yellow or red light that most nearly mimicked natural light was the most comforting to people. This makes sense because it is most natural and found within the organic lighting of the world. On the flip side, colors like blue and green light appeared to be the most cold to people and felt uninviting. Though these discoveries may seem surface level, they go deeper and can actually influence the emotions of people on a larger scale. These are important considerations to have when creating different preset lighting patterns/models for the ease of use of the teachers at Whitemarsh. If we would like there to be a lighting option that is welcoming and relaxing, we now have the research to back up a choice of making the lights a warm tone. The same goes for achieving other purposes. If we are teaching the children about the rain or cold weather, we could use blue lighting to intentionally elicit different feelings of discomfort or urge the students to do a particular action in that situation.

https://pmc.ncbi.nlm.nih.gov/articles/PMC9752890/

Electrical Engineering

According to Tufts University's engineering department:

- Forms of measurement : Lumens and watts
    - Lumens is a measure of light brightness over an area
    - Watt is a measure of energy consumption
      Light efficiency: brightness/energy
- The LED lights contain a circuit design, which is require because of their constant current
- LED lights also require a power supply, which could be a battery, outlet, ect
- Recombination- a current of electrons that flow along with a current that flows through the absence of electrons in the opposite direction the electrons and the holes recombine
    - Electrons are now in a higher energy level so as they fall down it emits the exact correct wavelength of light needed
    - Different LED colors are produced through the alteration in how far the electron falls into its corresponding hole



https://sites.tufts.edu/eeseniordesignhandbook/2015/leds-technology/

We need to measure the units of brightness of the LED lights so we can take into account the sensitivity of the students' eyes to the lights.

- The brightness of LED lights is measured in Lumens.
    - They measure the total amount of visible light emitted by the LED strips. The higher the number of lumens, the brighter the light.
    - This addressed the safety requirement that we have on the lights.

In creating our design, we will be incorporating many aspects of the engineering design process. For preliminary design and planning, we went through a complex process of considering stakeholder needs, design requirements, and specifications. This led to continued prototyping and planning for the design, as well as frequent communication with experts. This allowed us to practice our coding on a small scale version of our overall design. From there, we were able to begin creating and manufacturing an LED lighting system for the classroom that would properly achieve all designated functions. Throughout this entire process, we were able to incorporate mathematical and scientific thinking. Each iteration in our design came from recommendations from stakeholders and the different problems that came up along the way.

The design that we have created and plan on implementing at Whitemarsh Elementary's Room 3 fulfills all the needs and requirements, and it also follows general logic according to the research and opinions our group acquired from reliable sources. The design remains safe, reliable, and easy to use overall. Going with the simple design, we ensured that there was a level of satisfaction we as a group could guarantee for our stakeholders and that we knew we could live up to. Due to our consideration of different scientific and logistical elements like the effects that LED lights may have on the moods of children, we properly take into account all critical elements of the design. Therefore, our group is extremely confident in our design and believe that the process we used was reliable and ensures a stable final product that can be used for many more years.

Expert Review

1. **Mr. Muscarella**

Our first expert was Mr. Muscarella, a Plymouth Whitemarsh High School Engineering teacher. When we first approached him, we discussed why it would be important to make a design that was multifunctional and served all the purposes that our stakeholders needed. He helped us brainstorm how we could set up the lights in the room and what layouts would make the most sense. We talked about how we could code the lights and provide multiple functions for the room, while maintaining an easy to use control system.

As the project progressed, however, we realized that time and money were going to be a problem. Although we wanted to make a complex system of lights for the classroom, this was no longer realistic. Mr. Muscarella helped us realize that we could always complete this project in steps. The most important part of the project is that we are able to install lights in the room and that they are sectioned into zones with different color functions. Anything past that would simply be considered an "extra." As we planned out how to manage our time, he suggested that we install LED lights that come with a pre programmed remote. We can then use the built-in app in order to section off the lights and create different controls. Although this design would be less involved on our part, it still seemed like a viable idea. This also meant that we would be able to complete the project quickly and could always expand on it later and make some of our original ideas a reality.

2. **Dr. Sarver**

Dr. Sarver, a Professor at Drexel University (PhD in biomedical engineering), reviewed some of our preliminary ideas and design plans. We showed him our morphological charts, requirements, and options that we had created. As he examined our work, he highlighted some key components that we had not yet fully considered. Mainly, there was a lack of consideration of the students' needs in the classroom and how we hadn't planned enough adjustability in our designs. One thing he pointed out was our research on lumens. Lumens are a unit of measurement that tells us how much light is emitted by a lighting source. Although we had done some research on this topic, we hadn't considered the importance and implications that it had on our project. He pushed us to learn more about how bright was "too bright," and how different levels of brightness achieved different purposes. This meant that although we thought we were done with our research, we really did have to go back and re-examine if we had enough information to create a purposeful and intentional design.

As we did more research, we realized just how important it was to make the brightness of the LED lights adjustable for the class. Especially in a class where people have varying needs and circumstances, it became clear just how important it would be to give as many options as possible, and brightness would be a crucial part of this adjustability. We agreed as a group about how important it was to include this function as an option on the remote and made sure that this would be in our final design. Overall, this conversation helped inform decisions involving the adjustability/customization in brightness that we would provide for Room 3.

3. **Dr. Dougherty**

   We discussed our design with Dr. Dougherty, a professor at Drexel University's School of Biomedical Engineering, Science, and Health Systems with a PhD in Collaboration, and she told us that she liked our general design. She reminded us to cater to the needs of the classroom and to make sure to keep in mind the needs of the children. She has experience working with children, so she told us to factor in the kids' sensitivity to brightness and other triggers, like flashing. This was even more reason to make sure we included brightness controls in our design to prevent this from becoming an issue.

   She also told us to keep thinking of ideas to add to the lights, so that we can have a lot of options and keep our design unique. This encouraged us to focus on things like zoning, which she agreed would elevate our design and make it especially useful to the teachers at Whitemarsh.

   Dr. Doughtery's advice was most helpful in the sense that it helped us take a step back from our project and understand how the lights would function in the room and how useful they would really be. It allowed us to expand on some of our basic designs and include some extremely important features that would elevate the lights. Thanks to her, we were able to agree as a group to guarantee both brightness controls and zoning in any design we would propose to Whitemarsh.